A PROJECT TITLE

ON

Stock Return Prediction Using Ridge and Lasso Regression with Feature Scaling

MACHINE LEARNING
BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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Abstract:

Stock market prediction is a challenging task because stock prices and returns are influenced by various financial and market factors. This project focuses on predicting stock returns using **Ridge Regression and Lasso Regression** with appropriate **feature scaling** techniques. Historical stock data such as opening price, closing price, trading volume, high and low prices, and other relevant financial indicators are used as input features. Feature scaling is applied to bring all numerical features to a comparable range, improving the performance and stability of the regression models.

Ridge and Lasso Regression are regularized linear regression techniques that help reduce overfitting and handle relationships among multiple financial features. Ridge Regression controls model complexity by adding an L2 penalty, while Lasso Regression uses an L1 penalty and can reduce the importance of less useful features. The models are trained and evaluated using suitable performance metrics such as **Mean Squared Error (MSE), Mean Absolute Error (MAE), and R^2 score**. By comparing both approaches, the project aims to identify a suitable model for stock return prediction and demonstrate how feature scaling and regularization can improve machine learning-based financial prediction.

INTRODUCTION:

Stock market prediction is an important application of Machine Learning and Data Science that aims to analyze historical market data and predict future stock performance. Stock returns are affected by several factors, including opening price, closing price, high and low prices, trading volume, and market trends. Since these factors may have different numerical ranges and relationships with each other, directly applying a machine learning model can lead to inaccurate or unstable results. Therefore, feature scaling is used to transform the input features into a common scale before training the model.

This project, "Stock Return Prediction Using Ridge and Lasso Regression with Feature Scaling," focuses on using regularized regression techniques to predict stock returns. Ridge Regression uses L2 regularization to reduce the effect of large model coefficients, while Lasso Regression uses L1 regularization and can eliminate less important features. Feature scaling helps both models work effectively when the input variables have different ranges. The performance of the models can be compared using metrics such as Mean Squared Error (MSE), Mean Absolute Error (MAE), and R^2 score. The project demonstrates how regularization and feature scaling can be combined to build a simple and reliable machine learning approach for stock return prediction.

SOFTWARE REQUIREMENTS:

Programming Language: Python 3.x
IDE/Environment: Jupyter Notebook / Google Colab / VS Code
Libraries:
pandas — data handling and preprocessing
numpy — numerical computations and array operations
scikit-learn — Ridge Regression, Lasso Regression, StandardScaler, train-test split, and evaluation metrics
matplotlib — data visualization and plotting graphs
seaborn — statistical visualization and correlation analysis
Operating System: Windows/Linux/macOS (any OS supporting Python)
Hardware Requirements:
Processor: Intel i3 or above (i5 recommended)
RAM: Minimum 4 GB (8 GB recommended for smoother processing)
Storage: Minimum 500 MB free disk space
Display: Standard monitor (1366×768 or higher resolution)

Signature of the Guide **Course Instructor**